

KALENT CHIA CHANG RONG

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EDUCATION

National University of Singapore(NUS)

School of Computing (Business Analytics)

Faculty of Arts and Social Sciences (Economics)

Expected Aug 2028

GPA 4.68/5.00

WORK EXPERIENCE

Open Source Software Engineer, Mojaloop

May 2026 – Present

- Engineered TypeScript backend services and RESTful APIs for payment-testing infrastructure, ensuring reliability across multi-service Request-to-Pay workflows.
- Rebuilt error-response handling for the account-lookup microservice, tracing incorrect responses across multiple payment flows and restoring specification-compliant API behaviour.
- Modernized the central-ledger integration test pipeline by removing redundant reporting stages, eliminating recurring Node.js EPIPE failures and improving reliability by 10%.

Software Engineer Intern, DealersEdgeAsia

Jan 2026 – Jun 2026

- Led full-stack development of a scalable discovery and payments platform, designing PostgreSQL schemas for bookings and payments with Row-Level Security and React frontend components using Context API state management.
- Architected a payment workflow by integrating PayOS API with webhook processing and PostgreSQL RPCs, implementing signature verification and idempotency to prevent duplicate transactions for more than 200 users.
- Prototyped a LangGraph-based recommendation system for personalized merchant discovery, integrating user preferences with a PyTorch ranking model to generate tailored car-wash recommendations.
- Engineered a speech-processing pipeline using FFmpeg, Faster-Whisper and OpenAI APIs, parallelizing diarization across 4 concurrent threads and aligning speaker timestamps to reduce a 4 -day workflow to 5 hours.

PROJECTS

KittyWatch

April 2026

- Built a full-stack community cat tracking platform with Next.js, Supabase, implementing Redis rate limiting, geospatial queries and scheduled cron jobs for real-time sightings and disappearance detection.
- Engineered an image re-identification pipeline using local CLIP embeddings and pgvector cosine similarity, generating 512-dimensional image representations to verify sighting submissions against previously registered animals.

Finly

March 2026

- Architected a multi-service investment-analysis platform with React Native, FastAPI and Python, separating mobile, backend API and stateless agent-runtime services behind REST interfaces.
- Built a LangGraph-orchestrated 4-agent investment pipeline spanning research, fundamental, technical and risk analysis, integrating Exa, yfinance and Backtrader with OpenRouter-based access to OpenAI, Claude and Gemini models.

AWARDS

- NUS Merit Scholarship
- Agentic AI Build Week Hackathon Winner
- NUS Student Life Hackathon Top 8

SKILLS

Languages | Python, Java, JavaScript(ES2015+), TypeScript, C#, HTML, CSS, R, SQL

Frameworks | Node.js, Express, Vite, Next.js, React, React Native, Flask, Pandas, Playwright

Tools | Git, GitHub, MySQL, PostgreSQL, Docker, Figma, PyTorch, Redis, Firebase, Streamlit